Satellite Monitoring — Resolution & Cadence Cost Report

Prepared for: prospective project sponsor · **Prepared by:** Sophia Truesight (TrueSight DAO) · **Date:** 2026-09-11

1. Executive summary

- **No purchase is required to materially improve the satellite view.** A free upgrade path (Copernicus Data Space) raises rendered resolution and adds NDVI at **$0**.
- The paid step-up is small at our current scale: **Planet's 3 m near-daily agriculture imagery costs roughly $175–$900 per year** for our 273-hectare footprint.
- **50 cm on-demand tasking is the only five-figure option** (~$30k per collect) and is not appropriate for routine monitoring.
- **At the mission target of 10,000 hectares**, the same imagery ranges **~$3,500/yr (cheapest tier) to ~$18,000/yr (top tier)** — this is the number a sponsor should anchor to.

2. What the DAO runs today

Item	Current state
Source	Sentinel-2 L2A via Earth Search STAC — anonymous, free, no API key
Refresh	Daily cron 06:30 UTC
Cached scenes	4 per cell max, 45-day look-back
Grid	0.01° (~1 km) cells
Coverage	5 tree cells + 21 plot areas · 104 scenes · 4.4 MB total
Rendered asset	`preview.jpg` at 343 x 343 px (~10–49 KB)
Full tile (not stored)	241 MB per tile
Live basemap	Esri World Imagery, with our previews overlaid

Defect identified during this review: Esri World Imagery has no usable pixels at zoom 18 and above over the pilot plots (blank tile returned), so the basemap goes empty exactly at the zoom level needed to inspect individual canopies.

3. The two levers are independent

Lever	Free move	Paid move
Resolution	Request a larger render (512–1024 px) from the Copernicus Process API instead of the 343 px preview	PlanetScope at 3 m; then 50 cm tasking

Lever	Free move	Paid move
Cadence	Raise scenes-per-cell (4 to 8–12), shrink the grid, relax the cloud filter, add Landsat and Sentinel-1 (radar)	PlanetScope near-daily; on-demand tasking

4. Options and pricing

Option	Resolution / cadence	Cost	Fit for the DAO
Sentinel-2 (current)	10 m / ~5 days	$0	Baseline
Copernicus Data Space (Sentinel Hub)	10 m / ~5 days	Free tier 10,000 processing units/month; commercial units by quote	Free quality upgrade; on-the-fly true-color and NDVI
Sentinel-1 radar + Landsat 8/9	10 m / 30 m	$0	Sees through Amazon cloud cover
Planet Agriculture (per-hectare)	3 m / near-daily	$1.80 / $0.85 / $0.35 per ha/year (tiers 1/2/3), sold in 500 ha increments, annual commitment	Best value at our size
Planet Monitoring (area under management)	3 m / near-daily	$9,650/yr (50 km2 global); $2,700/yr (189 locations); $5,100/yr (49); $9,650/yr (17)	Overkill today
Planet Tropical Forest Observatory	4.77 m / monthly	Quote only (the free NICFI program ended April 2025)	Mission-aligned, tropical forest focus
Planet Nonprofit Program	As above	Tiered, quote only	The correct access door for us
SkySat 50 cm tasking	0.5 m / on demand	~$1,212 per km2, 25 km2 minimum (~$30k via Planet Select); resellers quote $15k tasking / $5k archive minimum	Not for routine monitoring

5. Cost at our current scale (273 hectares)

Tier	Rate	Cost per year
Tier 1	$1.80 / ha / yr	~$900
Tier 2	$0.85 / ha / yr	~$425
Tier 3	$0.35 / ha / yr	~$175

Note: imagery is sold in **500-hectare increments with an annual commitment**, so the practical entry price is set by that minimum, not by our 273 ha.

6. Cost at the mission target (10,000 hectares)

Tier	Rate	Cost per year
Tier 1	$1.80 / ha / yr	~$18,000
Tier 2	$0.85 / ha / yr	~$8,500
Tier 3	$0.35 / ha / yr	~$3,500

This is the figure a sponsor should plan against for full-scale verification of the 10,000-hectare Amazon rainforest goal.

7. What a sponsor's funds would buy

Investment	Deliverable
$0 (engineering time only)	512–1024 px true-color + NDVI tiles on the free Copernicus tier; fix the blank basemap at high zoom
~$175–$900 per year	3 m near-daily PlanetScope over all current plots — the resolution/cadence step-change
~$3,500 per year at scale	Same imagery across the full 10,000-hectare target
~$8,500–$18,000 per year at scale	Higher tiers with priority collection and support

8. Recommendation

1. **Free first.** Ship the Copernicus Process API renderer to replace the 343 px previews with 512–1024 px true-color and NDVI tiles on the free 10,000 PU/month tier, and fix the high-zoom basemap gap. This answers most resolution complaints at zero cost.

2. **Then quote.** Approach Planet for the Agriculture tier at the 500-hectare minimum plus the Nonprofit Program, and ask whether the Tropical Forest Observatory applies. Expect $175–$900/yr now, ~$3,500–$8,500/yr at scale.

3. **Exclude 50 cm tasking** from routine budgets — it is a per-collect product for one-off high-detail captures, not continuous monitoring.

9. Caveats

- Planet's rendered pricing page could not be scraped (bot-blocked). The per-hectare and area-under-management figures above come from published search results and should be treated as **indicative, not quoted**. All figures require a formal sales quote.
- The NICFI successor program status and pricing, and our eligibility for the Nonprofit Program, are quote-only.
- Unit economics are per-year and exclude engineering/labour to build and maintain the renderer.